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COMP1549 (2025/2026) - Advanced Programming

Security Access Control System (Task 2) - Project Reflection Report

1. Overview

This project implements an access-controlled system with users, roles, resources, access rules, typed capabilities using Java Generics, and logging of authorization decisions. Authorization uses a rule-chain approach, logging uses a decorator, and generics enforce correct usage at compile time.

2. Java Generics

Design Pattern Name	Classes/Methods	Generics Usage	Justification
Typed Capabilities	Capability, Read, Write, SecureResourceService	Capability<Read>, Capability<Write>	Ensures correct operation usage at compile time and prevents misuse.
Generic Access Outcome	AccessOutcome<T>, SecureResourceService	AccessOutcome<String>, AccessOutcome<Void>	Ensures correct handling of return types and prevents logical errors.

3. Design Patterns

Pattern	Classes/Methods	Justification
Chain of Responsibility	RuleChainAccessAuthorizer, AccessRule	Processes rules sequentially; default is REFUSE for security.
Decorator	LoggingAccessAuthorizer, AccessLog	Separates logging from business logic for clean design.

4. JUnit Testing

Test	Classes/Methods	Description
studentAllowedReadInternal	SecureResourceService.read	Ensures STUDENT can read

		INTERNAL resources.
studentDeniedWriteConfidentialAndLogged	SecureResourceService.write	Ensures STUDENT cannot write CONFIDENTIAL and logs REFUSE.
guestDeniedReadConfidentialAndLoggedFormat	AccessLogEntry	Ensures GUEST denial and correct log format.
adminAllowsReadWriteConfidential	SecureResourceService	Ensures ADMIN full access and logging.

5. Access Control Policy

Rule	Classes/Methods	Description
GUEST	GuestAccessRule	Can READ only PUBLIC resources.
STUDENT Read	StudentAccessRule	Can READ PUBLIC and INTERNAL.
STUDENT Write	StudentAccessRule	Cannot WRITE any resource.
ADMIN	AdminAccessRule	Full READ/WRITE access.

6. Usage of AI

AI Tool	Usage	URL	Contribution
GitHub Copilot	Documentation only	https://github.com/features/copilot	20% code generation

7. Presentation Style

This report follows the required structure with proper tables and justification. It uses Times New Roman font size 11 and is kept concise within 3 pages.